TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 6

5-MINUTE REVISION

Measures of Dispersion

Range, interquartile range, mean absolute deviation, variance, standard deviation and the coefficient of variation.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Report spread alongside the average.

REMEMBER

Variance has squared units; standard deviation has the original units. CV is useful only when a meaningful positive mean exists.

APPLY IT

Two crops can have the same mean yield and different variability. A smaller SD indicates more consistent yields on the same scale.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 What is dispersion?

The **extent to which observations deviate from a central value**: the spread of the data.

Absolute measures	Relative measures
In the data's units: range, IQR, MAD, SD	Unit-free: coefficient of variation (CV)

A good measure is rigidly defined, easy to compute, based on all values, suitable for further algebra and not unduly affected by extremes.

2 Range & IQR

Range

$$R = L - S$$

Interquartile range

$$IQR = Q_3 - Q_1$$

- **Range**: simplest; uses only the two extremes.
- **IQR**: spread of the **middle 50%**; less affected by extremes.
- **Quartile deviation** = IQR ÷ 2.

3 Mean absolute deviation

MAD about the mean

$$MAD = \frac{\sum |x_i - \bar{x}|}{n}$$

Uses all observations but ignores signs. MAD is **least about the median**.

4 Variance & standard deviation

Sample variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

Shortcut

$$s^2 = \frac{1}{n-1}\left\{\sum$$

Discrete frequency

$$s^2 = \frac{1}{n-1}\left\{\sum f_i x_i^2 - \frac{(\sum f_i x_i)^2}{n}\right\}$$

Grouped, $d_i = (x_i - A)/c$

$$s = c\sqrt{\frac{1}{n-1}\left\{\sum$$

SD s = non-negative square root of s^2. Grouped data: x_i = class mark, A = class mark with the highest frequency, c = class interval.

5 Properties to remember

- ✓ Variance is in **squared** units; SD is in the **original** units.
- ✓ SD is always ≥ 0; SD = 0 means all observations are identical.
- ✓ Sample variance uses divisor $n - 1$ (unbiased).
- ✓ Adding a constant to every value does **not** change SD.
- ✓ Multiplying by b multiplies SD by $|b|$.
- ✓ Squaring gives extreme deviations more weight.

6 Coefficient of variation

CV (%)

$$CV = \frac{s}{\bar{x}} \times 100$$

A **relative** measure, used to compare variability of datasets in **different units or with different means**.

Higher CV	Lower CV
More relative variability: less consistent	Less relative variability: more consistent

7 Comparing the measures

Measure	Uses	Extreme values	Note
Range	2 extremes only	**Strongly affected**	Quick, rough
IQR	Middle 50%	**Resistant**	Pairs with the median
MAD	All values	Moderate	Ignores signs
SD	All values	**Sensitive (squared)**	Best for algebra
CV	SD and mean	**Sensitive**	Unit-free; needs a positive mean

! Common mistakes

- ✗ Dividing by n for a sample variance.
- ✗ Reporting variance in the original units.
- ✗ Comparing SDs of data in different units (use CV).
- ✗ Forgetting to multiply by c in the grouped shortcut.

? Fill in the blanks

1. Variety A has SD 4 q and mean 40 q; variety B has SD 3 q and mean 20 q. The more consistent variety has CV = ____ %. **[10 (variety A)]**
2. If every observation is doubled, the SD is multiplied by ____ and the variance by ____. **[2; 4]**
3. If yield is recorded in kg, its variance is expressed in ____. **[kg^2]**
4. The observations 5, 5, 5, 5 have a standard deviation of ____. **[0]**
5. With Q_1 = 18 and Q_3 = 30, the quartile deviation is ____. **[6]**
6. Adding 10 kg to every observation changes the SD by ____. **[nothing (0)]**